Predict Customer Personality to boost marketing campaign by using Machine Learning

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Data-driven Information Systems Graduate with a strong foundation in Data Science, Web Development, and Database Management. Proficient in extracting insights using Python (Pandas, NumPy) and SQL. Experienced in managing large datasets and optimizing operational workflows through analytical approaches. Currently enhancing expertise in Machine Learning and Predictive Modeling through an intensive Data Science Bootcamp. Proven ability to adapt quickly to new technologies and deliver actionable insights for business growth

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“A company can grow rapidly when it understands the behavior of its customer personalities, enabling it to provide better services and benefits to customers with the potential to become loyal customers. By processing historical marketing campaign data to improve performance and target the right customers to transact on the company’s platform, the focus is on building a cluster prediction model to help the company make better decisions.”

Background, Problem Statement & Objective

Background

- Companies struggle to identify which customers are most likely to respond to marketing campaigns.
- Sending campaigns to all customers wastes budget and lowers ROI.
- Dataset: 2,240 customers, 29 features — demographics, spending behaviour, and campaign history.

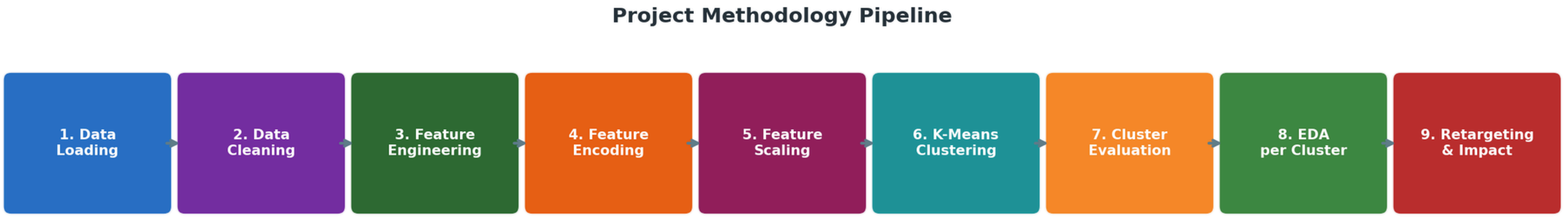
Problem Statement

How can we use machine learning to identify distinct customer personality segments, and which segments should be prioritised for marketing retargeting to maximise campaign ROI?

Project Objectives

1. Perform feature engineering (conversion rate, age groups, spending features)
2. Conduct EDA to understand customer behaviour and campaign effectiveness
3. Clean data: handle missing values, duplicates, and outliers
4. Apply feature encoding and standardisation for ML readiness
5. Determine optimal number of clusters using Elbow Method
6. Implement K-Means clustering and evaluate with Silhouette Score
7. Interpret cluster profiles and identify retargeting candidates
8. Calculate potential business impact of targeted marketing

Methodology — Project Pipeline



Dataset

2,240 customers

29 original features

Target

Response (0/1)

Campaign acceptance

Ref Year

2014

Age calculation base

Final Rows

2,212

After cleaning (98.75%)

Key Features Engineered:

ConversionRate — sum(Response)/sum(NumWebVisitsMonth) — segment-level

Age — 2014 − Year_Birth

AgeGroup — Bins: 18-29, 30-39, 40-49, 50-59, 60-69, 70+

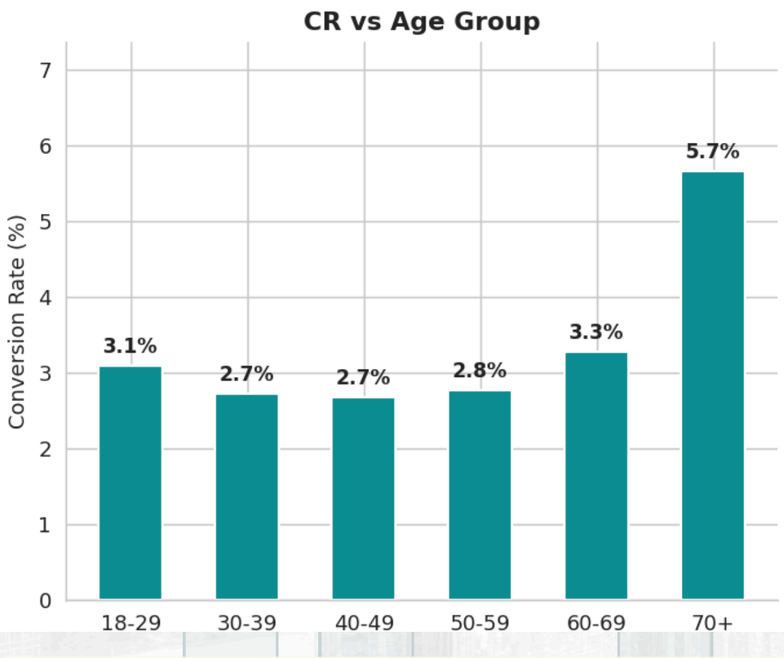
TotalSpend — Sum of 6 product spending columns

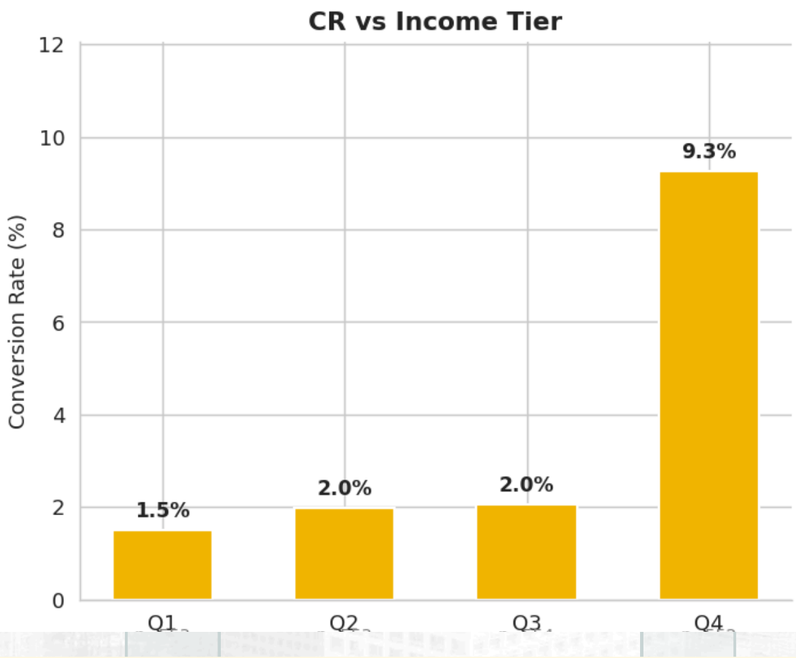
TotalTransactions — Sum of purchases across 4 channels

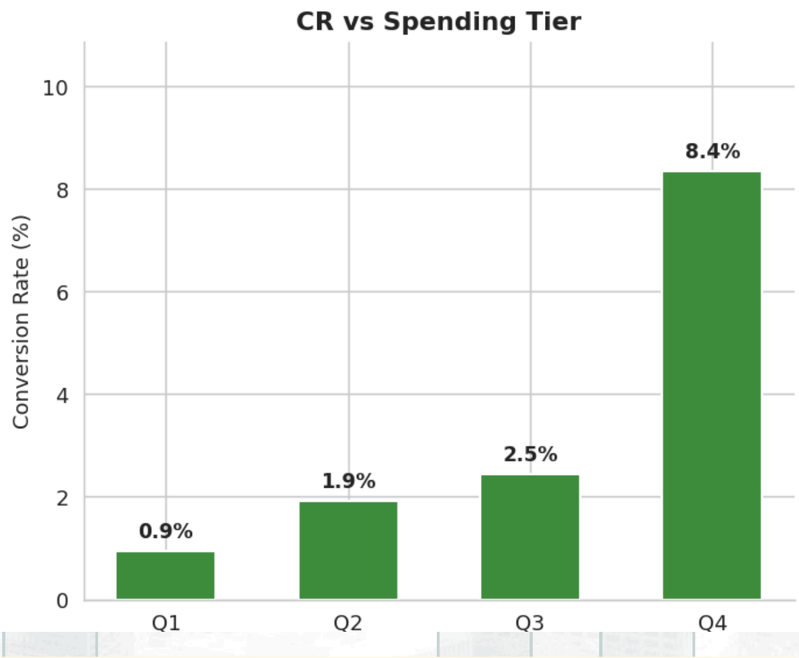
TotalChildren — Kidhome + Teenhome

IncomeTier / SpendTier — Quartile-based segmentation

Feature Engineering — Conversion Rate & Age Group Analysis







Interpretation & Statistical Test Results

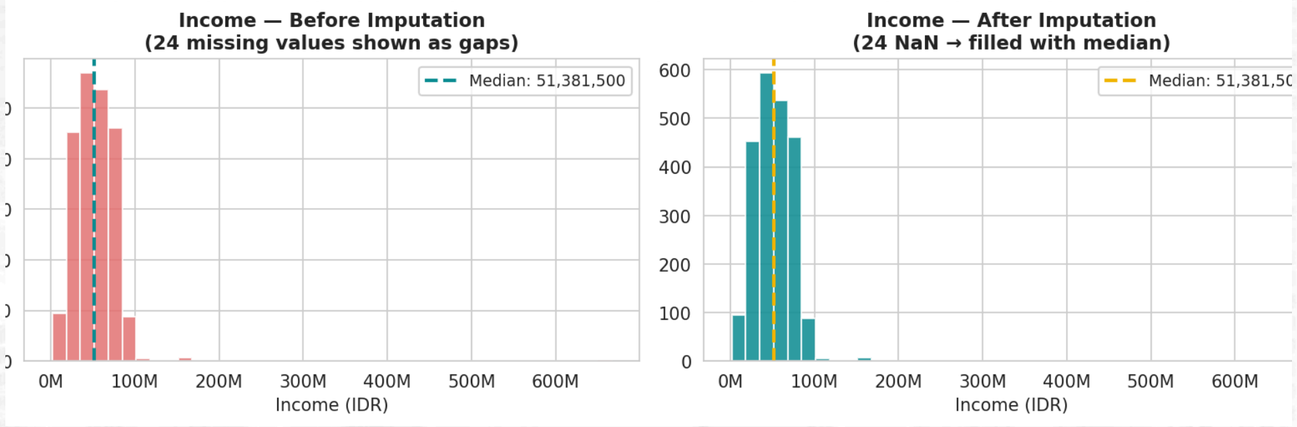
Variable	Conversion Rate Range	p-value	Significance
Age Group	2.7% – 5.7%	0.728	**Not Significant**
Income Tier	1.5% – 9.3%	< 0.0001	**Highly Significant**
Spend Tier	0.9% – 8.4%	< 0.0001	**Highly Significant**

Age is NOT a meaningful predictor — income & spending tier drive conversion rate.

Data Cleaning — Missing Values & Duplicated Rows

Missing Values

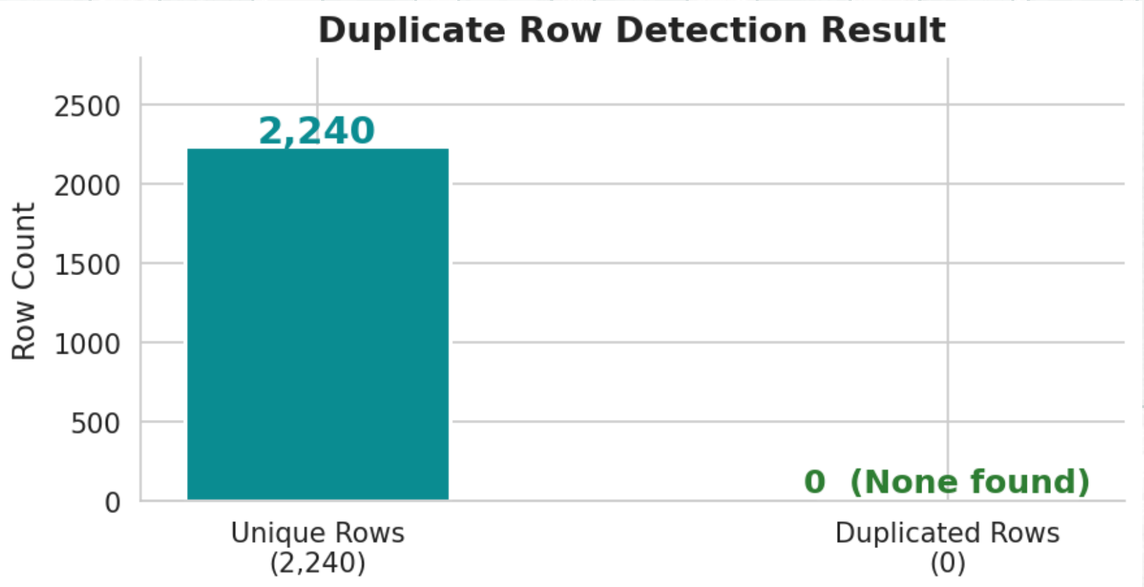
Column	Count	Missing %	Action
Income	24	1.07%	Imputed with median (right-skew robust)
All other columns	0	0%	No action needed



Duplicated Rows

0

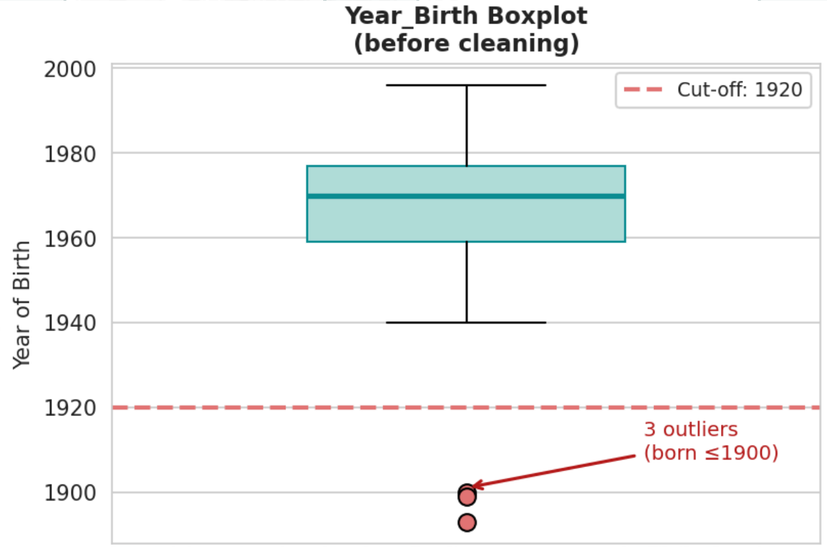
duplicate rows found
→ No removal required

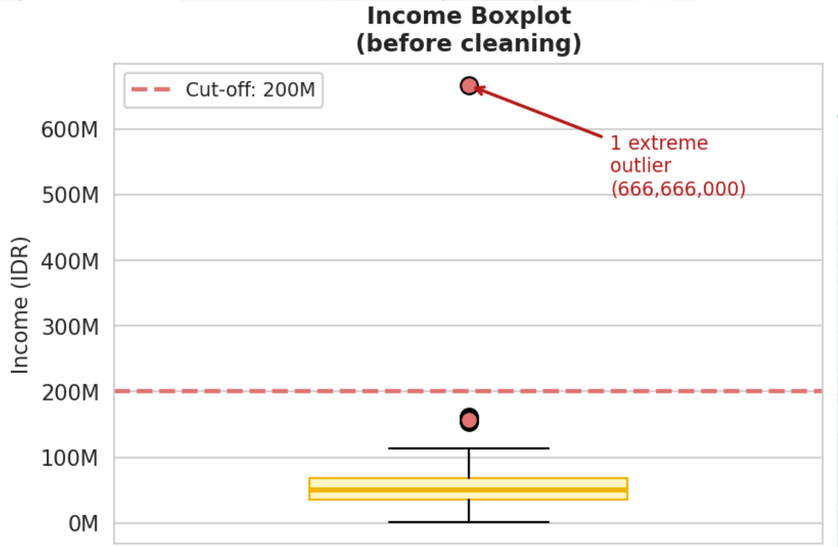


Check	Result	Status
df.duplicated().sum()	0	**PASS**
Rows retained	2,240 / 2,240	**100%**

Data Cleaning — Outlier Removal & Summary

Outlier Detection (Year_Birth & Income)





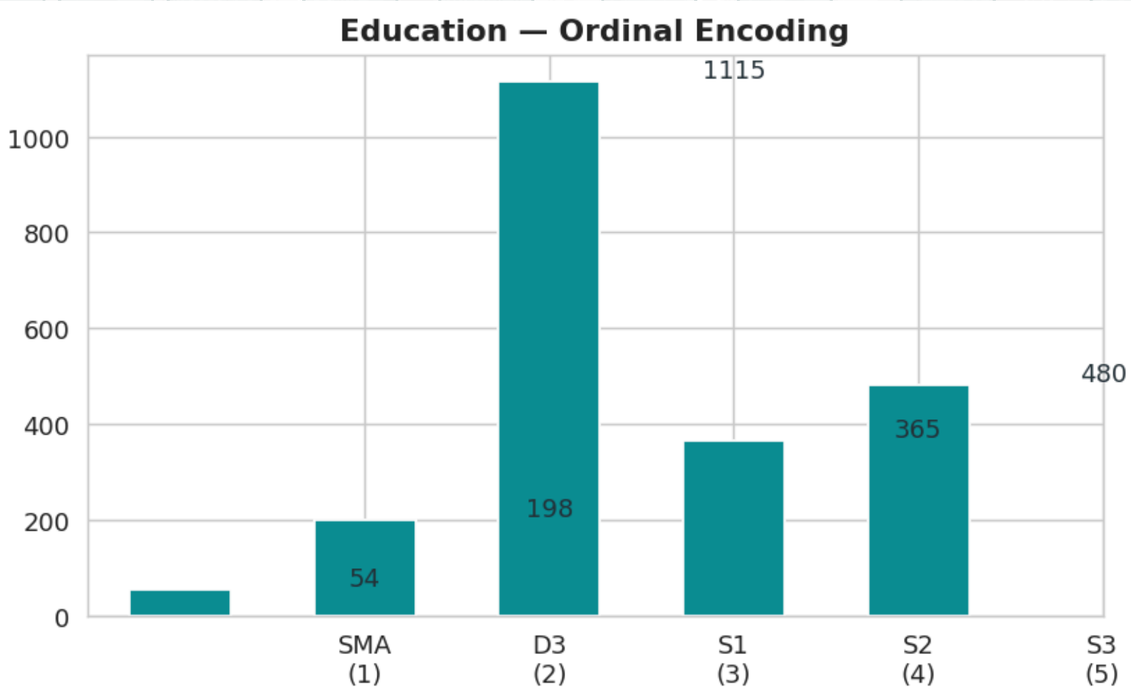
Cleaning Log — All Steps

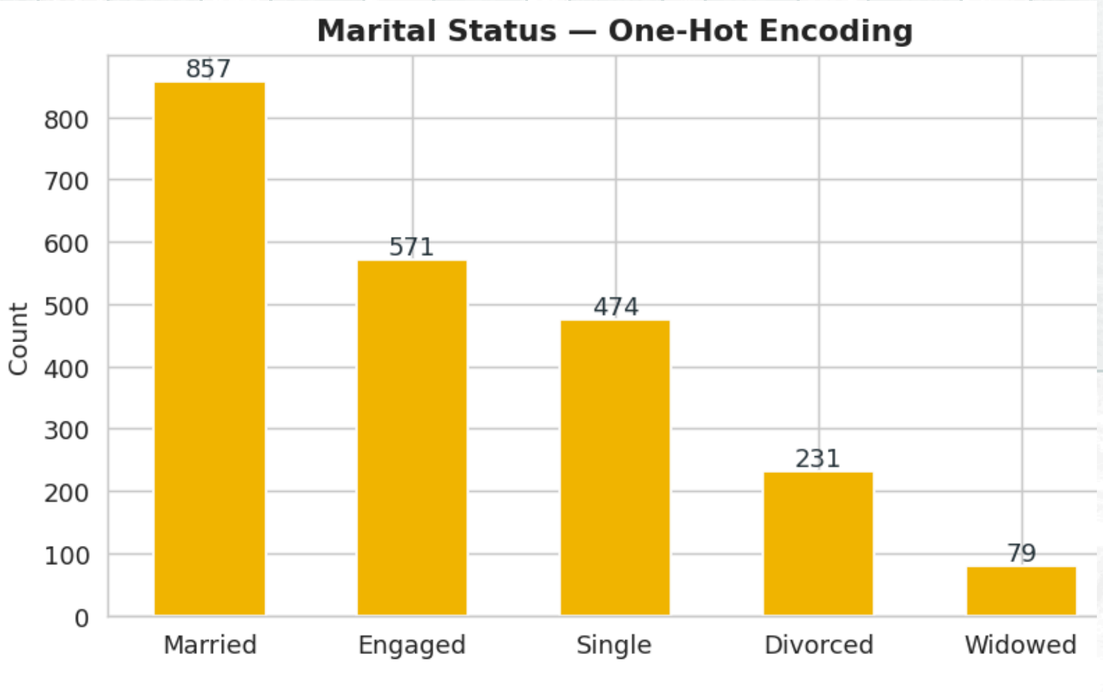
Step	Issue	Rows	Action	Result
1	Missing Income (NaN)	24	Imputed with median	0 missing
2	Year_Birth < 1920 (age > 94)	3	Dropped	Removed
3	Income > 200M (repeating digit)	1	Dropped	Removed
4	Duplicate rows	0	No action needed	N/A
Final	**2,212 of 2,240 rows retained**		**98.75% ✓**	

Data Preprocessing — Feature Encoding

Categorical columns converted to numeric form before modelling.

Column	Method	Rationale
Education	Ordinal Encoding (SMA=1 → S3=5)	Has a natural educational hierarchy
Marital_Status	One-Hot Encoding (5 binary columns)	Nominal — no inherent order between categories





Data Preprocessing — Feature Standardisation

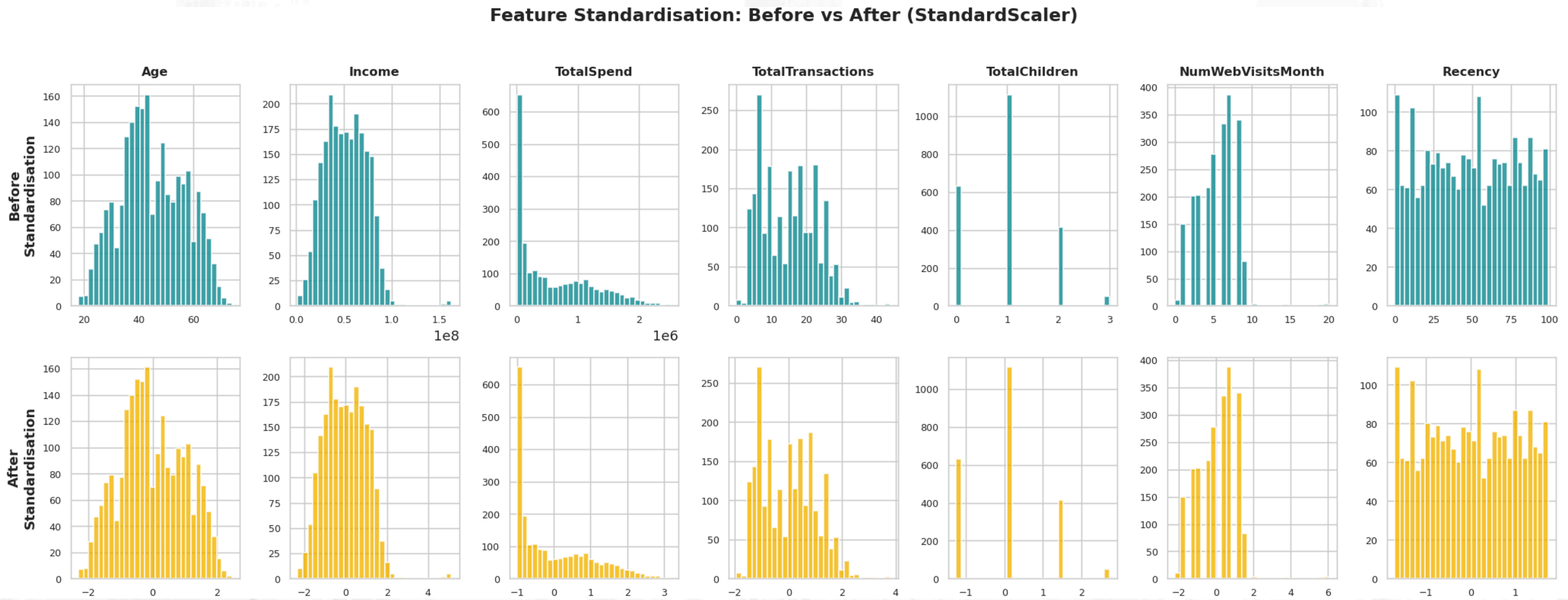
StandardScaler (Z-score) is applied to all numeric features so that high-scale variables (e.g. Income in millions) do not dominate the model. Formula: $z = (x - \mu) / \sigma$

Feature	Mean (before)	Std (before)	Mean (after)	Std (after)
Age	52.2 yrs	11.9	≈ 0	≈ 1
Income	52,247,015	21,859,180	≈ 0	≈ 1
TotalSpend	607,648	603,543	≈ 0	≈ 1
TotalTransactions	14.7	7.5	≈ 0	≈ 1
NumWebVisitsMonth	5.3	2.4	≈ 0	≈ 1
Recency	49.1	28.9	≈ 0	≈ 1

Why Standardise?

- Income (millions) vs Age (years) — huge scale difference
- K-Means uses Euclidean distance — sensitive to scale
- After scaling: mean ≈ 0, std ≈ 1 for all features

Data Preprocessing — Feature Standardisation



K-Means Clustering — Elbow Method & Silhouette Score

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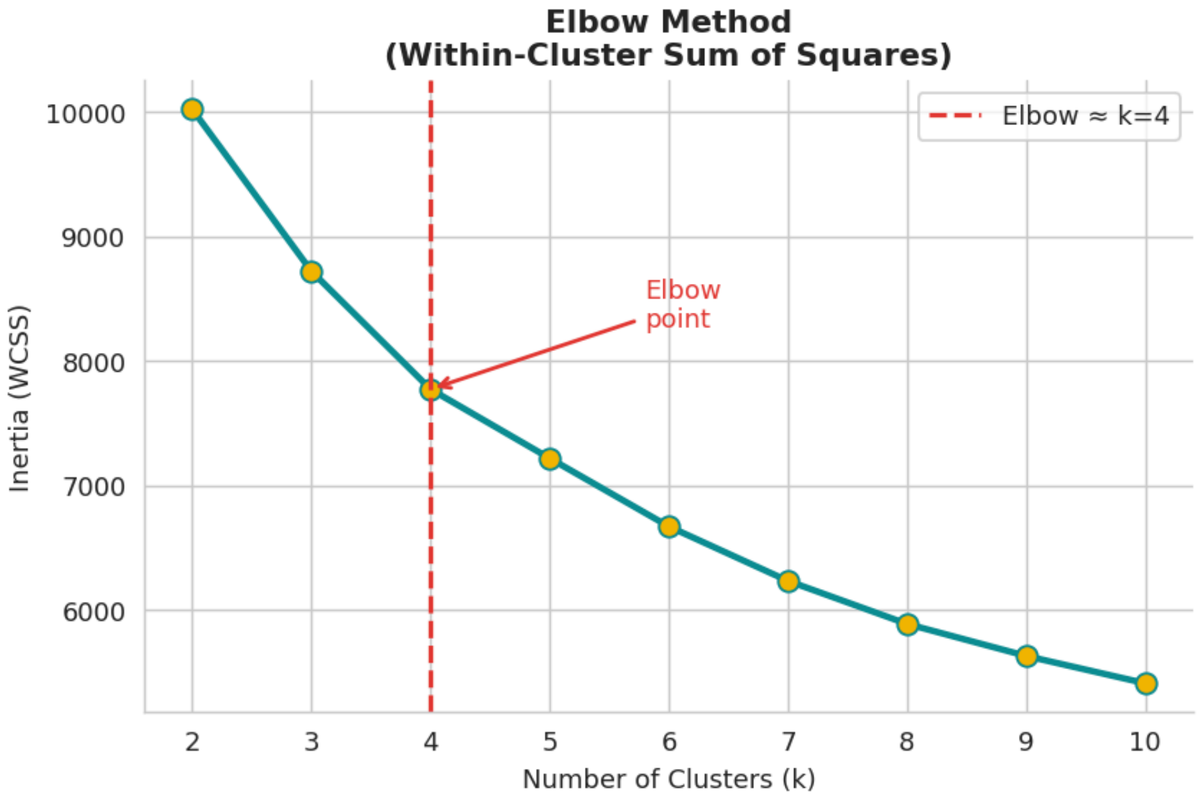
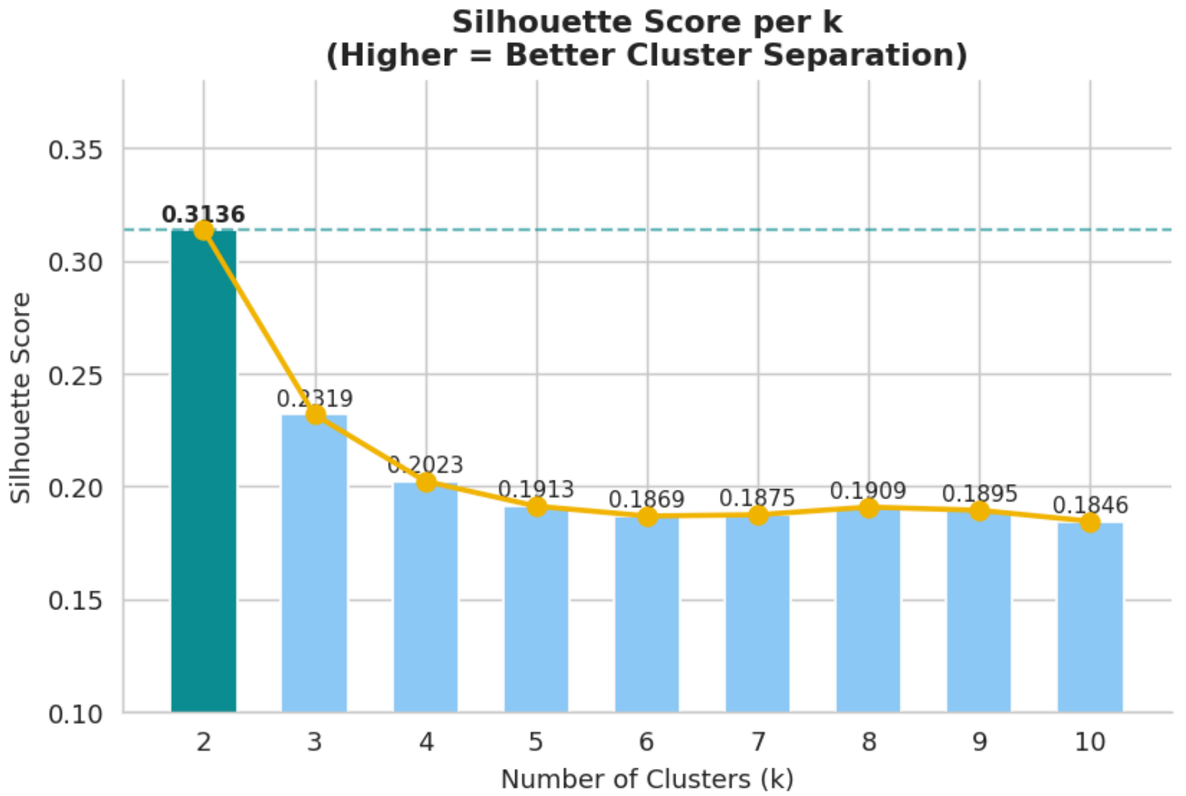
Features: Age · Income · TotalSpend · TotalTransactions · TotalChildren · NumWebVisitsMonth · Recency (all StandardScaler-normalised)





Elbow Point

k = 4

Inertia curve flattens — diminishing returns after k=4

Best Silhouette (k=2)

0.3136

k=2 too coarse — only 2 segments, limited business value

Selected k

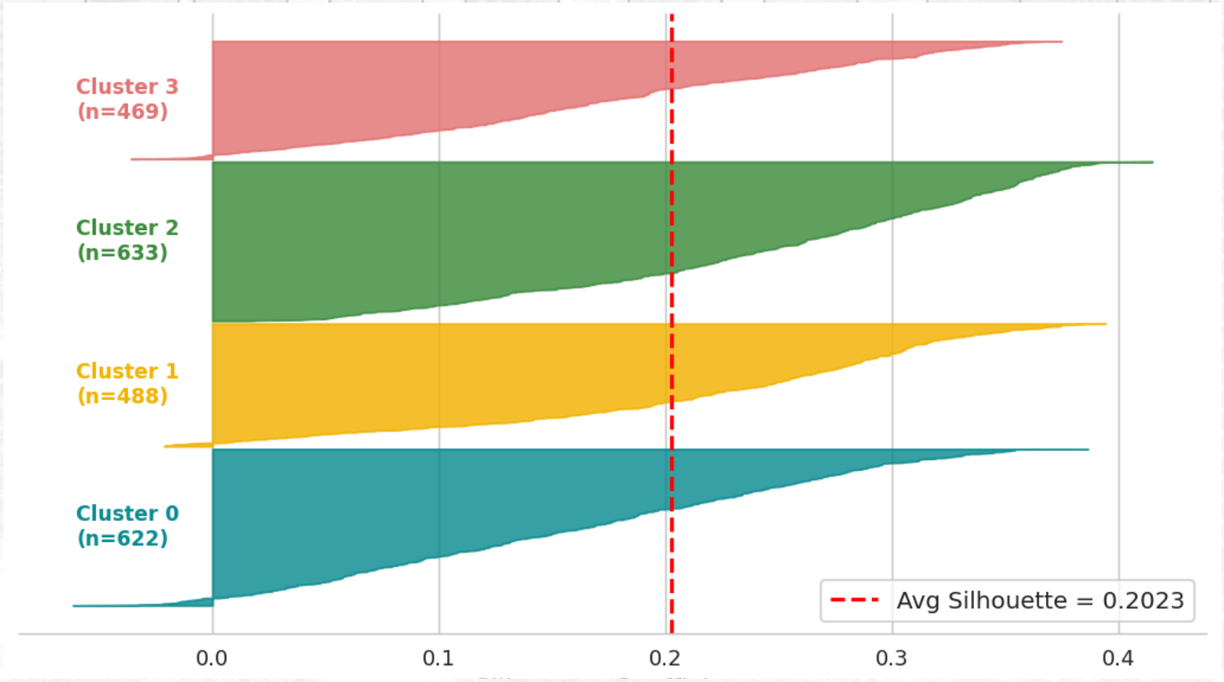
k = 4

Balances statistical fit (elbow) + 4 actionable segments

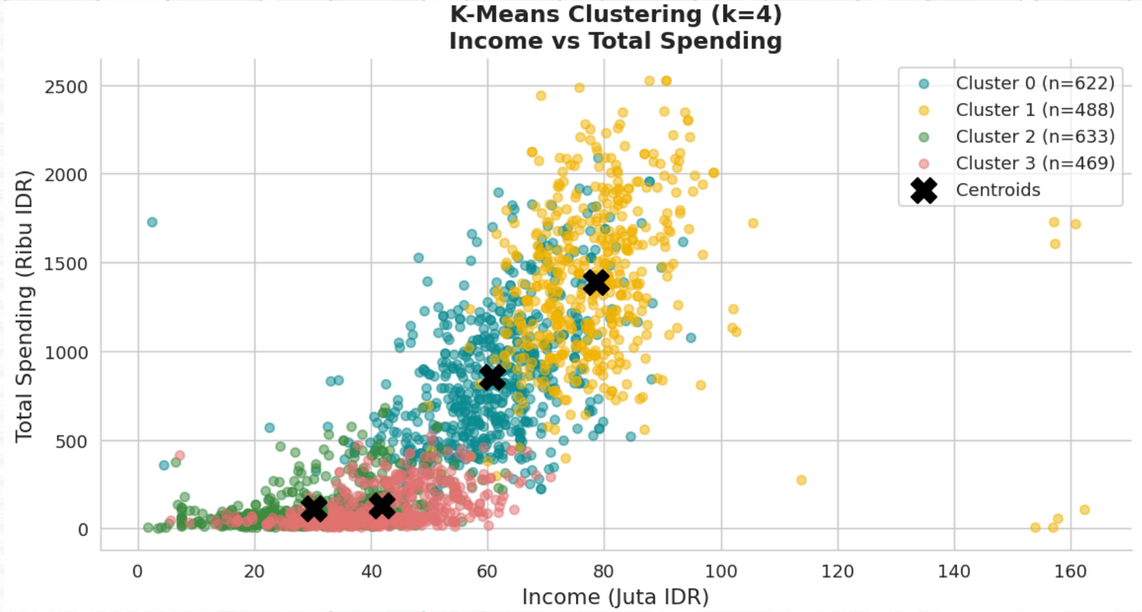
K-Means Clustering — Model Evaluation (k=4)

Model Evaluation — k=4

Metric	Value	Interpretation
Inertia (WCSS)	7,772	Clusters are internally compact
Silhouette Score	0.2023	Moderate separation — normal for multidim. customer data
Elbow Point	k = 4	Clear diminishing returns after k=4
Cluster Sizes	622 / 488 / 633 / 469	Reasonably balanced segments

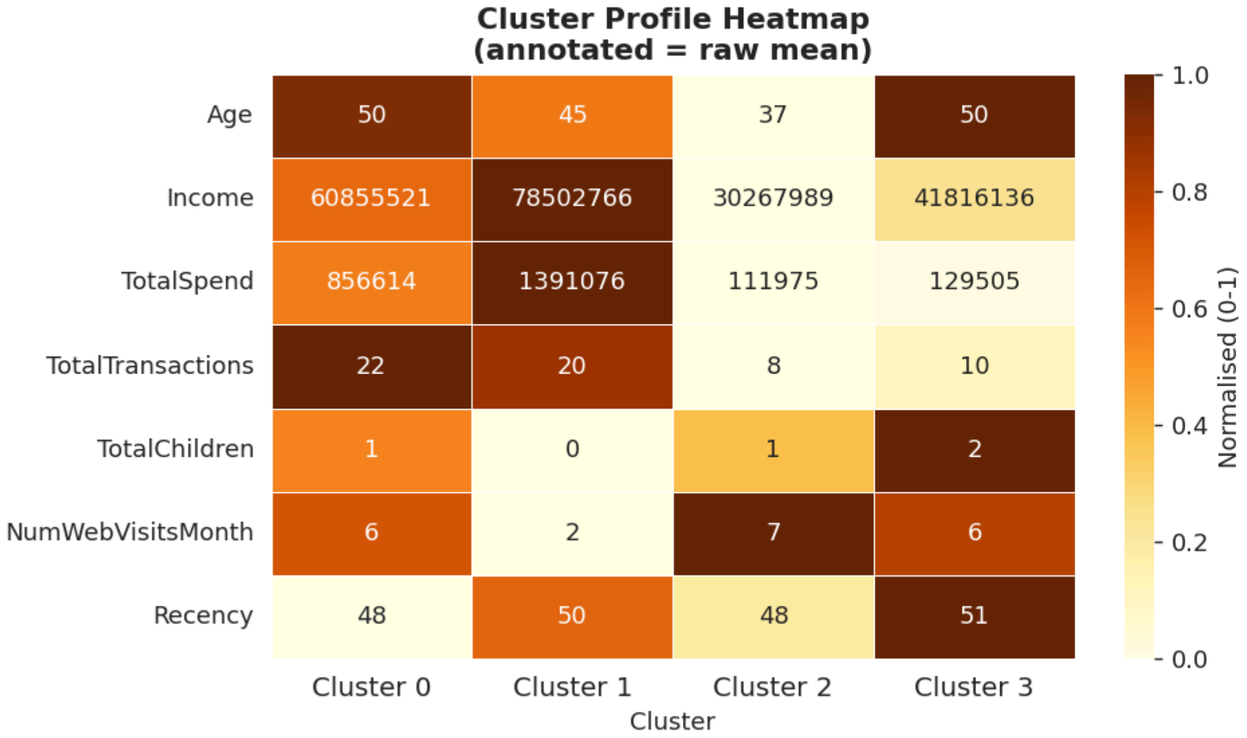


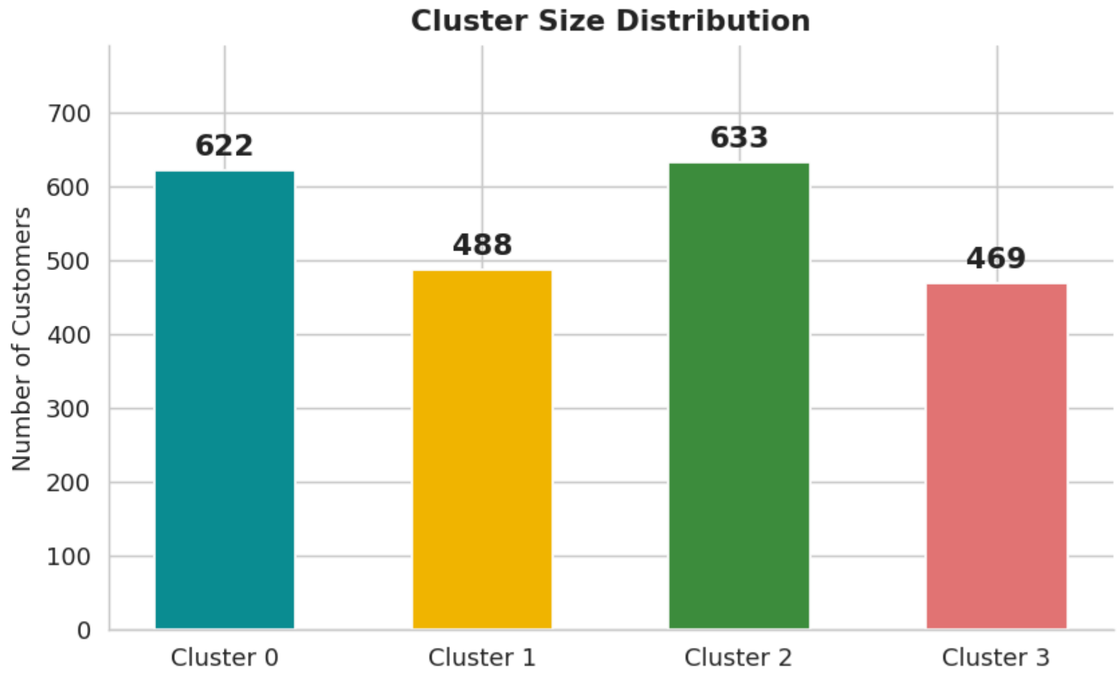
Cluster Profile & Distribution



K-Means Clustering — Model Evaluation (k=4)

Cluster Profile Heatmap and distribution





K-Means Clustering — Cluster Interpretation

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C0 — Mid-High Spender

```
Income: ~61 Jt
Spend: ~857 Rb
Age: ~50 y
Children: 1.1
```

Mature, moderately high spending. Good upsell and premium loyalty target.

C1 — Premium Buyer

```
Income: ~79 Jt
Spend: ~1.39 Jt
Age: ~45 y
Children: 0.1
```

Highest income & spend, nearly child-free. Most responsive — top priority.

C2 — Low Engagement

```
Income: ~30 Jt
Spend: ~112 Rb
Age: ~37 y
Visits: 6.9/mo
```

Low income, high web visits but low conversion. Funnel optimisation needed.

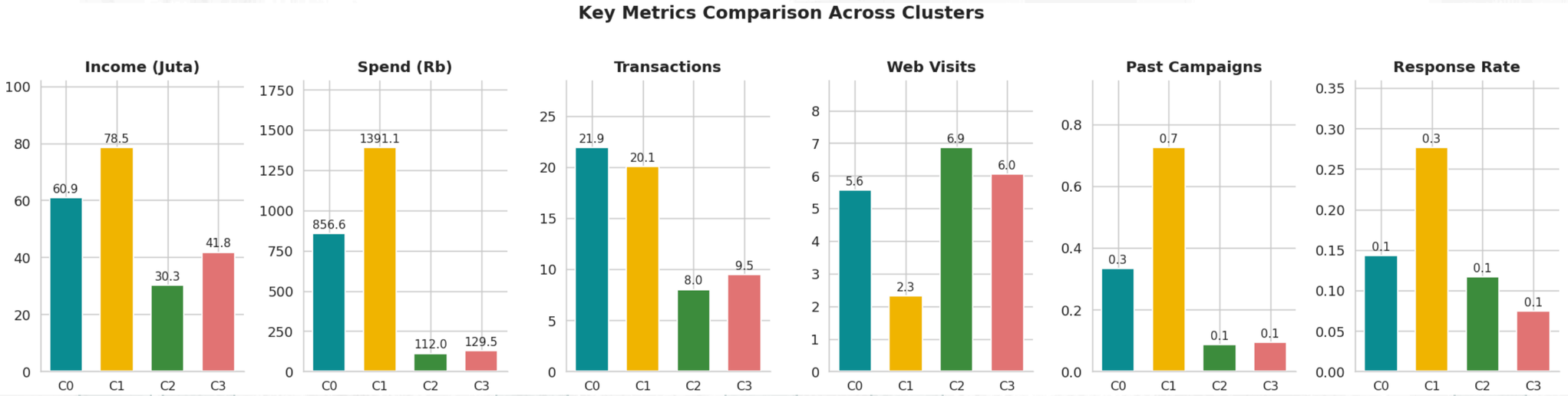
C3 — Family Budget

```
Income: ~42 Jt
Spend: ~130 Rb
Age: ~50 y
Children: 1.9
```

Most children, tightest budget. Family bundle promos and value messaging.

EDA Cluster Analysis — KPI Overview & Distribution

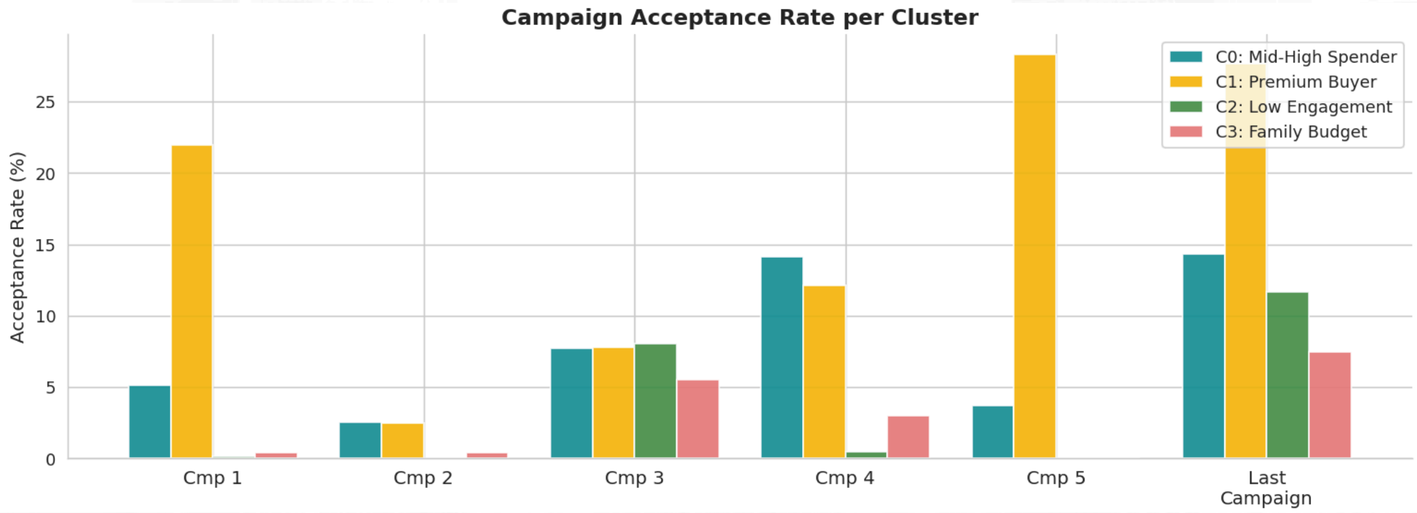
Key Metrics Comparison Across Clusters



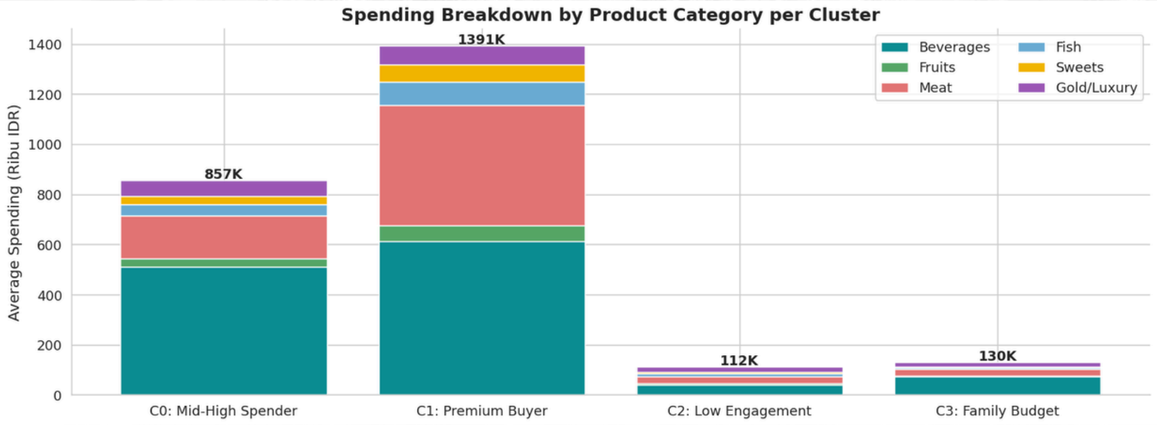
Income & Spending Distribution (Violin Plot)

EDA Cluster Analysis — Campaign, Spending & Channel Behaviour

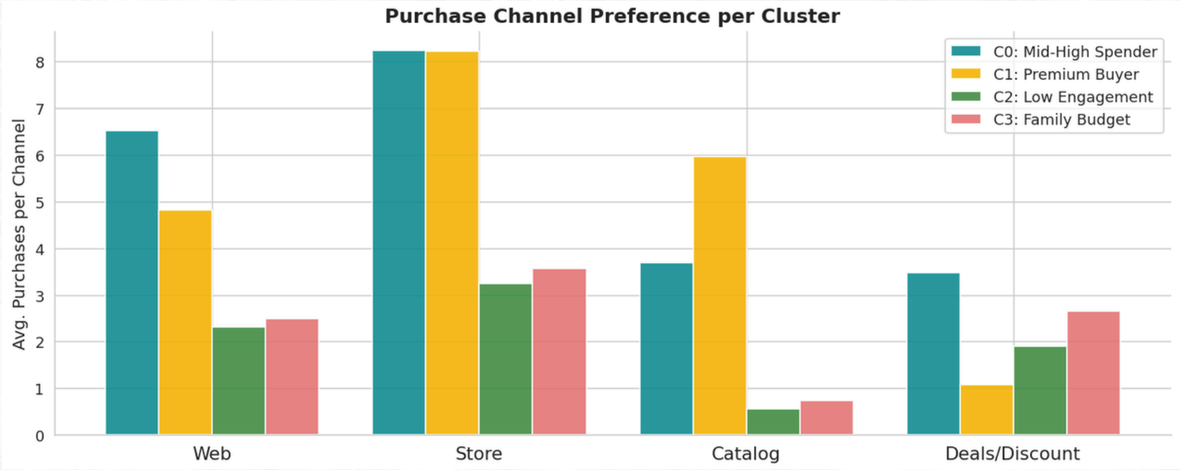
Campaign Acceptance Rate per Cluster



Spending by Product Category

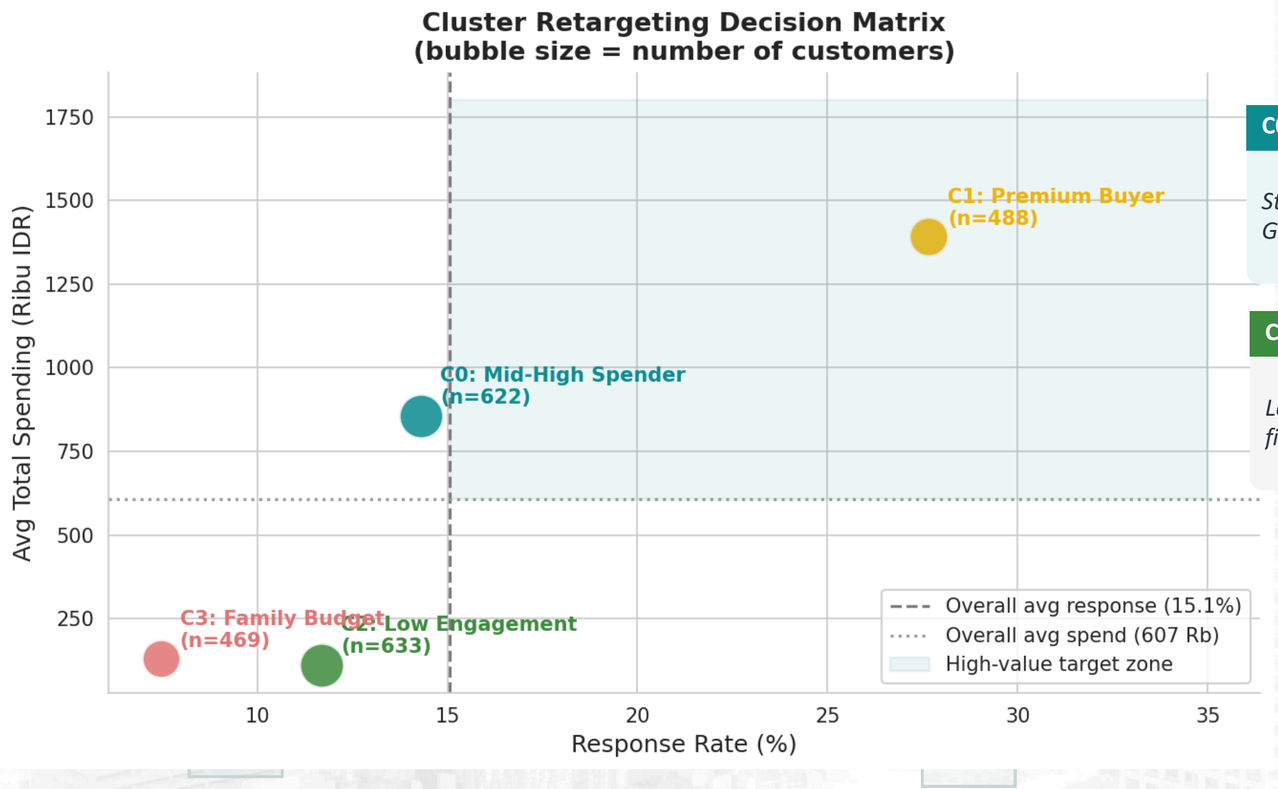


Purchase Channel Preference



Cluster Selection — Marketing Retargeting

Retargeting Decision Matrix (Response Rate vs Average Spending)



C0: SECONDARY TARGET

Strong response (14.3%) and high spending (857 Rb). Good upsell potential with loyalty programs.

C1: PRIMARY TARGET

Highest response rate (27.7%), highest spending (1.39 Jt), nearly child-free. Maximum campaign ROI.

C2: LOW PRIORITY

Low response despite high visits. Needs funnel fix first before retargeting investment.

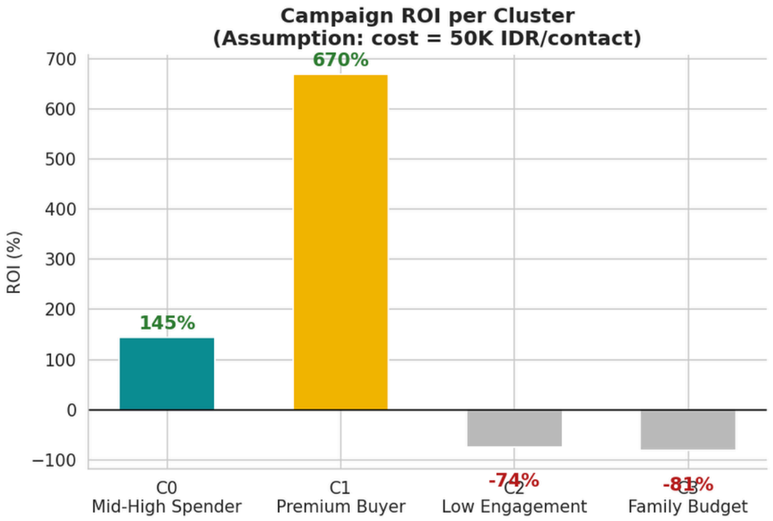
C3: LOW PRIORITY

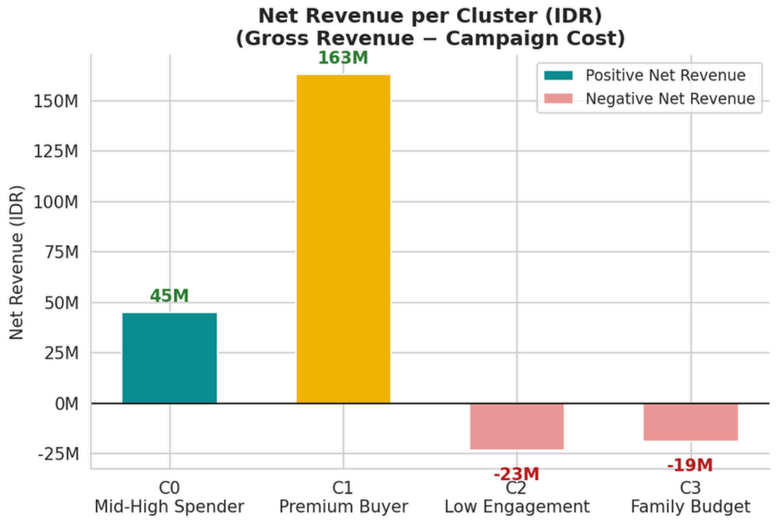
Lowest response rate (7.5%) and lowest average spending. Negative ROI at standard contact cost.

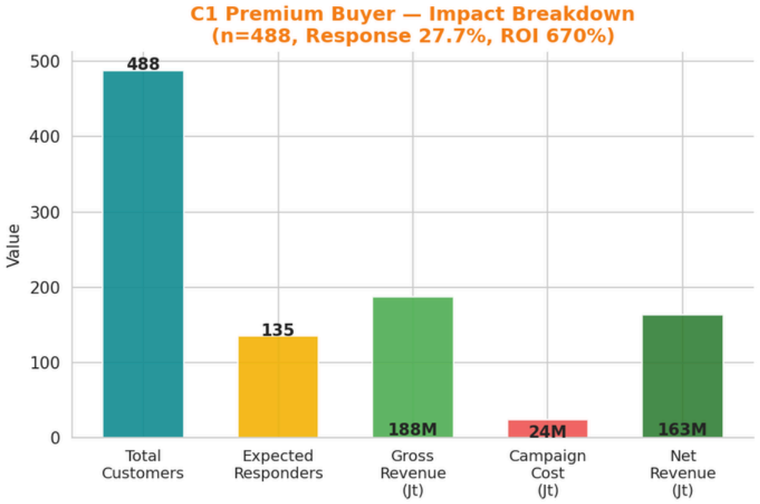
Selected Targets: C1 (Premium Buyer) + C0 (Mid-High Spender)

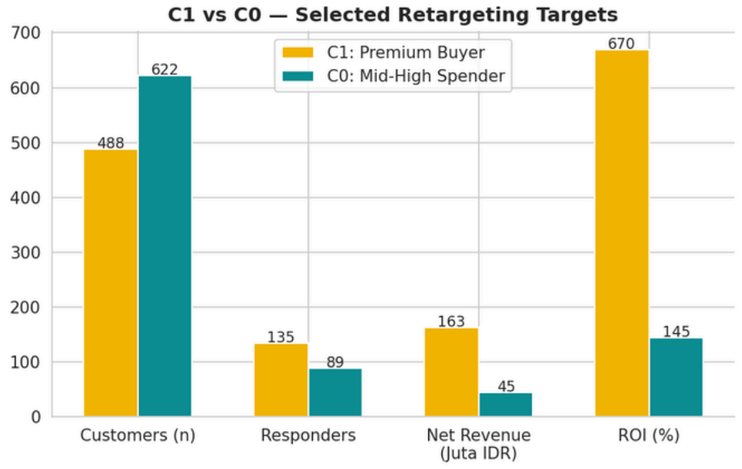
Combined Customers	Avg Response Rate	Expected Conversions	Combined Campaign Cost
1,110	**21.1%**	**224**	**55.5 Juta**

Potential Impact — Marketing Retargeting Analysis









Cluster	n	Response Rate	Gross Revenue	Campaign Cost	Net Revenue	ROI
C1 Premium	488	27.7%	187.8 Jt	24.4 Jt	**163.4 Jt**	**670%**
C0 Mid-High	622	14.3%	76.2 Jt	31.1 Jt	**45.1 Jt**	**145%**
C2 Low Engage	633	11.7%	8.3 Jt	31.7 Jt	**-23.4 Jt**	**-74%**
C3 Family	469	7.5%	4.5 Jt	23.5 Jt	**-18.9 Jt**	**-81%**
TOTAL (C1+C0)	1,110	21.1%	264 Jt	55.5 Jt	**208.5 Jt**	**376%**

Business Recommendations Based on Cluster Analysis

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C0 — Mid-High Spender

Secondary

Income: ~61 Jt
Spend: ~857 Rb
Transactions: 22
Response: 14.3%

Best Channel: Web + Store

- Upsell to premium product tiers
- Loyalty rewards & membership
- Omnichannel nudge (web + store)
- Beverage & Gold/Luxury cross-sell

C1 — Premium Buyer

PRIMARY ★

Income: ~79 Jt
Spend: ~1.39 Jt
Transactions: 20
Response: 27.7%

Best Channel: Catalog + Campaign

- Top priority for ALL campaigns
- Personalised catalog offers
- Exclusive premium bundles
- High-value retention program

C2 — Low Engagement

Retention

Income: ~30 Jt
Visits: 6.9/mo
Spend: ~112 Rb
Response: 11.7%

Best Channel: Web Re-engagement

- Fix web conversion funnel
- Drip email re-engagement
- Entry-level product recs
- Incentive first-purchase discount

C3 — Family Budget

Activation

Income: ~42 Jt
Children: 1.9
Spend: ~130 Rb
Response: 7.5%

Best Channel: Deals / Promo

- Family bundle & combo deals
- Value-for-money messaging
- Discount / cashback promos
- Kids & family product push

Budget Allocation:

- C1 Premium — 40%
- C0 Mid-High — 30%
- C2 Low Engage — 18%
- C3 Family — 12%

Source Code

All analysis in this presentation is fully reproducible from the notebook below:

`conversion_rate_analysis.ipynb`

https://drive.google.com/file/d/1psDJiTwVbaBkxxMhOvA9nSkKBHI_azsK/view?usp=sharing

Complete Pipeline Coverage:

01 **Data Loading & Exploration —** Shape, dtypes, sample review

02 **Data Cleaning —** Missing values (24 Income NaN), outliers (4 rows), duplicates (0)

03 **Feature Engineering —** Age, AgeGroup, TotalSpend, TotalTransactions, ConversionRate, IncomeTier, SpendTier

04 **Feature Encoding —** Ordinal (Education) + One-Hot (Marital_Status)

05 **Feature Standardisation —** StandardScaler Z-score on 7 numeric features

06 **EDA — Conversion Rate —** Analysis by Age, Income, Spending with significance testing

07 **K-Means Clustering —** Elbow Method (k=2–10), final model k=4, Silhouette Score 0.20

08 **Cluster EDA —** KPI comparison, campaign acceptance, spending breakdown, channel preference

09 **Retargeting & Impact —** Cluster selection, ROI calculation, net revenue per cluster